

Q. What is Activity? How Activity works? Life cycle of Activity?

Like C, C++ java has some entry point i.e. main() method, android has also its entry point i.e. onCreate() method which is present in Activity class.

⇒ Activity is one of the most important core building block of Android OS.

⇒ Activity is just a simple class which represents a single screen.

⇒ An activity is one screen of an app. Activity is similar to a window in the windows OS.

⇒ The most specific block of the user interface is the activity.
EX:

Login screen, sign up screen & home screen.

How does Activity work?

⇒ An activity in Android is a specific combination of XML files and Java files.

⇒ It is basically a container that contains the design as well as coding stuff.

⇒ XML files provide the design of the screen and Java files deal with all coding stuff like handlers, what is happening, design files etc.

⇒ JAVA files and XML files make activity complete.

⇒ To make an app activities are needed that are bound loosely with each other.

⇒ In the Android manifest the activities can be organized in parent-child relationships to aid navigation.

⇒ The "main activity" is called the first activity.

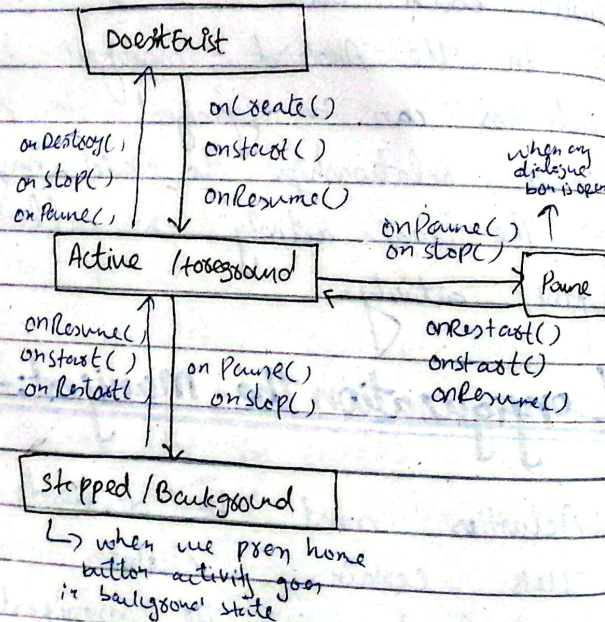
Configuration the manifest:-

⇒ Activities must be declared and there certain attributes must be declared in the manifest so

that activities used in our app
 ⇒ Android manifest is a file
 that give information about the
 application.

State/Phases of Activity:

- (1) Active state
- (2) Pause state
- (3) Stopped state
- (4) Destroy state



Q. Describe the Activity Life Cycle?

⇒ In android, an activity is referred to as one screen in an application. It is very similar to a single window of any desktop application.

⇒ An android app consists of one or more screens or activities.

⇒ Each activity goes through various stages of a life cycle and is managed by activity stacks.

⇒ So when a new activity starts, the previous one always remains below it.

⇒ These are four stages of an activity:-

- (1) if an activity is in the foreground of the screen i.e. at the top of the stack, then it is said to be active or running. This is usually

the activity that the user is currently interacting with.

(2) If an activity has lost focus and a non-full-sized or transparent activity has focused on top of your activity.

o In such a case either another activity has a higher position in multi-window mode or the activity itself is not focusable in the current window mode. Such activity is completely alive.

(3) If an activity is completely hidden by another activity, it is stopped or hidden. It still remains retains all the information and as its window is hidden thus it will often be killed by the system when memory is needed elsewhere.

(4) The system can destroy the activity from memory by either asking it to finish or simply killing its process. When it displayed again to the user, it must be completely restarted and restored to its previous state.

⇒ For each stage, android provides us with a set of 7 methods that have their own significance for each stage in the life cycle.

1) onCreate:

⇒ It is called when the activity is first created. This is where all the static work is done like creating views, binding data to lists, etc. This method also provides a Bundle containing its previous frozen state, if there was one.

2: OnStart()

⇒ It is invoked when the activity is visible to the user. It is followed by `onResume()` if the activity is invoked from the background. It is also invoked after `onCreate()` when the activity is first started.

3: onResume()

⇒ It is invoked after the activity has been stopped and prior to its starting stage and thus is always followed by `onStart()` when any activity is revived from the background to on-screen.

3: onPause()

⇒ It is invoked when the activity starts interacting with the user. At this point, the activity is at the top of the activity stack, with a user interacting with it.

Always followed by `onPause()` when the activity goes into the background or is closed by the user.

5: onPause(): onStop():

⇒ It is invoked when an activity is going into the background but has not yet been killed.

It is a counter part to `onResume()` when an activity is launched in front of another activity, this callback will be invoked on the top activity.

⇒ The activity, under the active activity's `onP` will not be created until the active activity's `onPause()` returns, so it is recommended that heavy processing should not be done in this part.

6: onRestart()

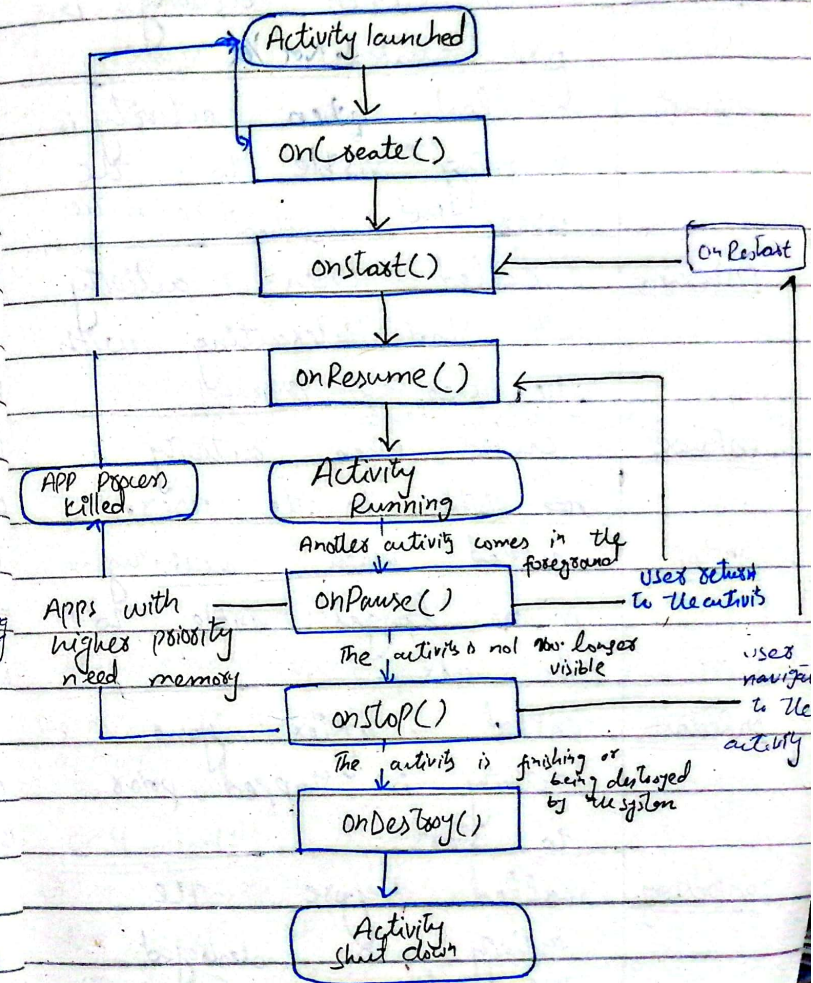
⇒ It is invoked when the activity is not visible to the user. It is followed by `onRestart()` when the activity is revoked from the background, followed by `onDestroy()` when the activity is closed or finished, and nothing when the activity remains on the background only.

⇒ This method may never be called, in low memory situations where the system does not have enough memory to keep the activity's process running after its `onPause()` method is called.

7: onDestroy()

⇒ The final call received before the activity is destroyed. This can happen either because the activity is finished or

because the system is temporarily destroying this instance of the activity to save space.

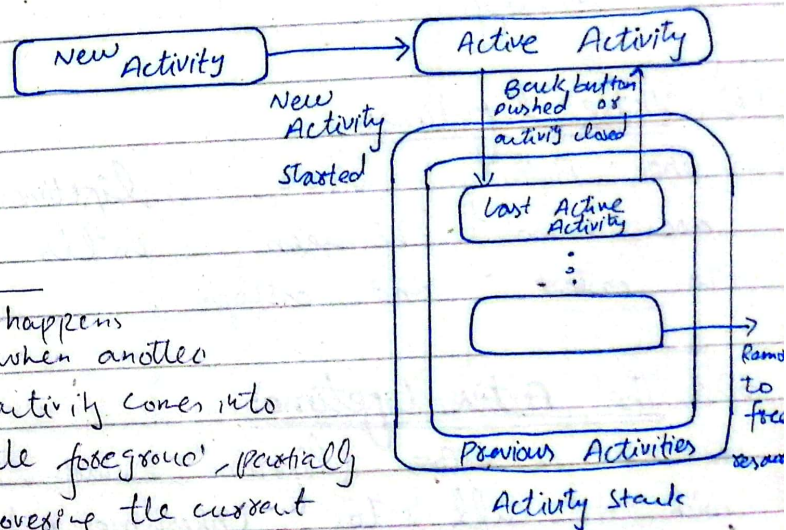


Summary:

method	Description
<u>onCreate</u>	called when activity is first created. Not yet visible.
<u>onStart</u>	called when activity is becoming visible to the user. But it may not be in the foreground.
<u>onResume</u>	called when activity will start interacting with the user in the foreground.
<u>onPause</u>	called when activity is still not visible to the user. ←
<u>onStop</u>	called when activity is no longer visible to the user.
<u>onRestart</u>	called after your activity is stopped prior to start.
<u>onDestroy</u>	called before the activity is destroyed. - when activity is removed from memory.

Q. what is activity stacks?

⇒ The state of each activity is determined by its position on the Activity stack, a last-in-first-out collection of all the currently running activities.



happens when another activity comes into the foreground, partially covering the current activity.

onStopped

→ activity is no longer visible to the user.
- stopped due to the user navigating to another activity or if the system needs resource.

Activity Lifetimes

(1) Full Lifetime:

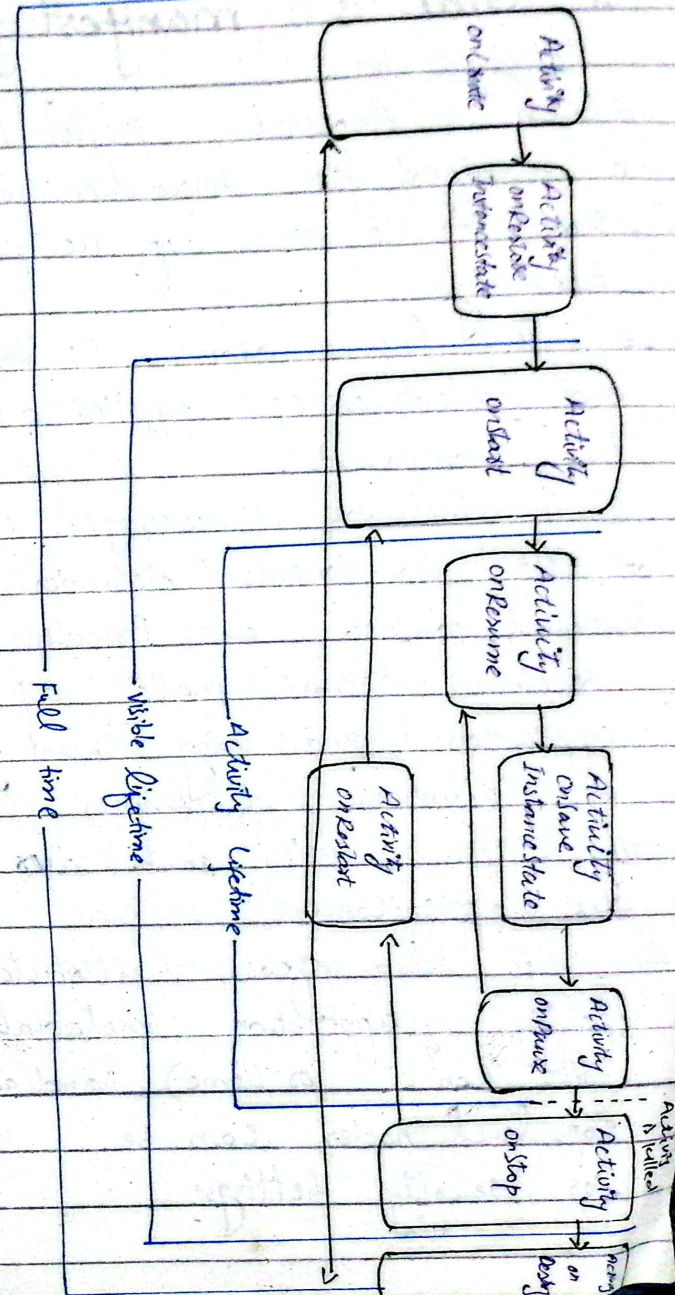
⇒ The full lifetime of activity occurs between the first call to onCreate and the final call to onDestroy.

(2) Visible Lifetime:-

⇒ The Activity's visible lifetimes are bound between calls to onStart and onStop.

(3) The Active Lifetime:

⇒ The active lifetime starts with a call to onResume and ends with a corresponding call to onPause.



Topic:

Android Activity Classes

⇒ The android SDK includes a selection of Activity subclasses that wrap up the use of common user interface widgets. Some of the more useful:

MapActivity:

- Encapsulates the resource handling required to support a MapView widget within an Activity.

ListActivity:

- Wrapper class for Activities that feature a ListView bound to a data source as the primary UI metaphor, and exposing handlers for list item selection.

ExpandableListActivity:

Similar to the List activity.

Activity Group:

- Allows to embed multiple activities within a single screen.